

# Non-Generative Learning to Rerank for MS/MS-Based Molecule Retrieval

**Abstract.** Retrieving molecular structures from tandem mass spectra is difficult because many candidates share mass, formula, and substructure information. Joint embedding methods rank by spectrum-molecule similarity, whereas generative retrieval may introduce an autoregressive molecular decoder. We study direct, non-generative learning to rerank a retrieved molecular list. SpecEmbedding-Rerank first aligns a peak-sequence spectrum encoder with a molecular graph encoder. Its second-stage residual ranker combines spectrum and candidate embeddings, explicit interaction features, the base score, and the base rank under listwise supervision. Using the overlap-clean alignment-seed-42 checkpoint and caches, we compare a pointwise scorer without learned cross-candidate interaction with a candidate-set Transformer across three reranker training seeds; a separate analysis repeats the design for alignment seeds 42-44. On the 17,556-query MassSpecGym test split, the fixed-alignment mean  $\pm$  sample-SD Recall@1 values are  $67.44 \pm 0.74$  and  $68.58 \pm 0.20$  for mass-conditioned retrieval, and  $73.97 \pm 0.59$  and  $74.57 \pm 0.28$  for formula-conditioned retrieval, compared with fixed-base values of 47.46 and 63.10. Recall and MRR use supplied candidate files with a local exact-target-SMILES single-positive protocol; the unquantified difference from reference two-dimensional InChIKey identity, which may yield multiple positives, means they are not equivalent to official evaluator outputs. Across the three alignment seeds, both learned rankers improve their corresponding base in all 12 alignment-candidate-learned-model aggregate cells for both Recall@1 and MRR, whereas the Transformer-pointwise direction varies by alignment. A fixed-alignment audit finds that the full model does not achieve six-of-six MRR wins against any of five removal variants; the no-rank-embedding variant improves both metrics in every pool-by-reranker-seed pair. Thus, the evidence supports supervised non-generative second-stage reranking; robust self-attention and independent component benefits remain unestablished. Neither variant generates molecules or forces the target into the test top-40 list.

**Keywords:** Data mining for bioinformatics · Tandem mass spectrometry · Molecule retrieval · Learning to rank · Multimodal representation learning

## 1 Introduction

MS/MS is widely used to identify small molecules in metabolomics, natural-product discovery, and related biomedical applications. An observed spectrum is only a partial, condition-dependent view of a molecular structure: different

molecules can yield similar fragments, while the same molecule can produce different spectra across instruments, adducts, and collision energies. Consequently, structure annotation is naturally a large-scale data mining problem in which a query spectrum must retrieve and rank a correct structure from a chemically constrained candidate pool.

MassSpecGym provides structure-disjoint data splits and standardized mass- and formula-conditioned candidate pools for this problem [3]. Recent methods mainly address the cross-modal gap between spectra and molecules. JESTR learns a spectrum-molecule joint space with contrastive multiview coding and ranks candidates directly by cosine similarity [13,21]. GLMR instead retrieves contextual candidates, generates a refined molecule with a conditional language model, and reranks candidates by molecule-molecule similarity [25].

Both approaches leave an important data mining question open. Candidate-wise cosine scoring does not explicitly train a second-stage listwise objective over the retrieved set, while generative retrieval adds an autoregressive decoder and makes ranking depend on the quality and validity of a generated structure. We ask whether a supervised second-stage ranker can directly separate hard candidates and whether explicit candidate interaction adds value beyond a pointwise residual scorer.

Building on the peak-sequence spectrum architecture introduced by SpecEmbedding [23], we construct SpecEmbedding-Rerank, a two-stage retrieval and reranking framework. The first stage trains that spectrum tower from scratch and aligns it with a GINE molecular graph encoder. The second stage operates only on the first-stage top- $K$  list. It constructs explicit spectrum-candidate interaction features, injects the original score and rank as retrieval priors, and predicts a residual correction to the base score. We instantiate this framework with pointwise and candidate-set Transformer variants.

Our contributions are:

- We instantiate and systematically audit direct, non-generative second-stage learning-to-rank for MS/MS-to-molecule retrieval, avoiding both cosine-only final ranking and autoregressive molecular decoding.
- We compare pointwise and self-attentive rankers with matched features, score heads, objectives, and optimization. On the same fixed caches, we then audit five removal settings of the full configuration.
- Using the MassSpecGym split and candidate files under our local exact-target-SMILES single-positive protocol, both implementations improve their fixed base retriever across alignment seeds 42–44. The candidate-set Transformer advantage is not directionally uniform across these checkpoints, so the analysis does not establish a robust self-attention benefit. Moreover, within the overlap-clean alignment-seed-42 audit, the full model does not achieve six-of-six pool-by-reranker-seed MRR wins against any removal variant.

*AI-assistance disclosure.* OpenAI Codex assisted throughout all sections with drafting and language revision, and during the project with code editing, experiment orchestration, and consistency audits. Reported observations and numerical results came from the stated software pipeline, not model generation. The

authors reviewed all AI-assisted text and code, verified the technical claims, citations, saved artifacts, and reported results, and assume full responsibility for the manuscript.

## 2 Related Work

### 2.1 Annotation Paradigms

Computational MS/MS annotation differs in retrieved objects and ranking evidence. Candidate-score correction predates our work: MetFusion combines in-silico fragmentation with spectral-library evidence [8]; MS2Query reranks its top 2,000 MS2DeepScore matches with a five-feature random forest [12]; and LC-MS2Struct combines an MS/MS scorer with retention order to jointly rank candidates from one LC analysis [1]. We instead rank structures in query-specific MassSpecGym pools from frozen spectrum-molecule representations, without spectral-library-neighbor or retention-time evidence [3].

Within structure-database retrieval, CSI:FingerID and MIST predict molecular fingerprints from spectra [7,9]. CMSSP and JESTR learn joint spectrum-molecule spaces; MVP extends contrastive alignment across molecular graphs, fingerprints, individual spectra, and consensus spectra [5,13,6]. These methods order candidates by predicted-fingerprint scores or cross-modal similarity. SpecEmbedding instead studies spectrum-spectrum representation learning [23]; we reuse only its peak-sequence architecture, train it from scratch against molecular graphs, and learn a complementary second-stage order within a retrieved hard-candidate list.

Other paradigms use different intermediate objects. MassFormer predicts tandem spectra from molecular graphs [24], while MSNovelist generates structures from MS/MS-derived fingerprints [20]. GLMR conditions its molecular decoder on retrieved candidates and reranks them by similarity to the generated molecule [25]. Our closed-set method does not decode a structure or return a molecule outside the first-stage list; it directly optimizes the scores of available candidates.

### 2.2 Learning to Rank Candidate Sets

Learning-to-rank objectives define supervision over preferences or complete lists rather than treating similarity sorting as a fixed post-processing step. RankNet learns pairwise preferences, whereas ListNet defines a probability model over a ranked list [2,4]. Ranking functions can also exploit relations among competing items. Set Transformer provides attention blocks for set-structured inputs [15], while SetRank uses self-attention to model cross-document interactions [17]. We apply these principles to cached cross-modal embeddings with explicit spectrum-candidate pair features and first-stage score and rank priors. Table 1 positions the resulting query-specific, non-generative second stage among selected related MassSpecGym approaches. The contribution is neither reranking nor self-attention itself; residual scoring and candidate interaction are evaluated design choices rather than assumed sources of gain.

**Table 1.** Mechanistic comparison with selected related MassSpecGym approaches.

| Model | Ranking signal                                         | Candidate interaction                        | Generates | Training stages                                      |
|-------|--------------------------------------------------------|----------------------------------------------|-----------|------------------------------------------------------|
| JESTR | Spectrum-candidate cosine                              | None at inference                            | No        | InfoNCE plus formula-candidate regularizer           |
| GLMR  | Generated-candidate cosine                             | Retrieved candidates condition the generator | Yes       | Dual-path InfoNCE plus generation                    |
| Ours  | Residual score from pair features and retrieval priors | Optional candidate self-attention            | No        | Cross-modal InfoNCE plus listwise/pairwise reranking |

### 3 Problem Formulation

Let  $s$  be an MS/MS spectrum,  $m^+$  its correct molecule, and  $\mathcal{D}_s$  a query-specific mass- or formula-conditioned candidate pool. A base retriever assigns

$$b(s, m) = \text{sim}(f_s(s), f_m(m)) \quad (1)$$

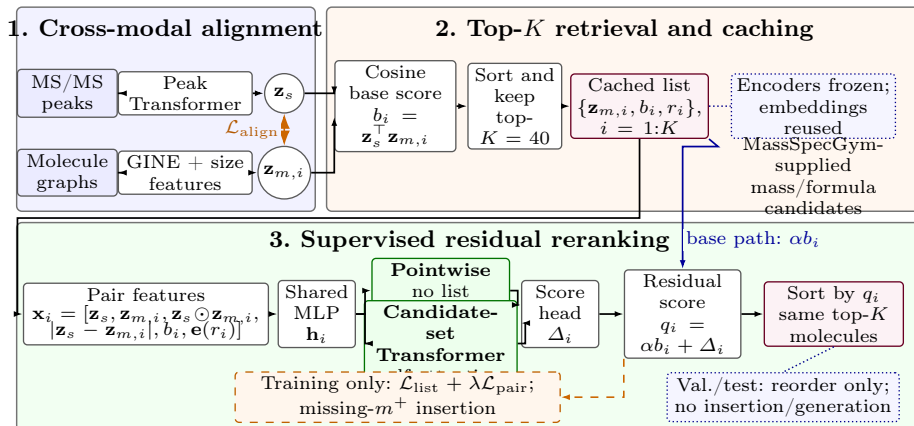
to every  $m \in \mathcal{D}_s$  and returns the top- $K$  list  $\mathcal{C}_s = (m_1, \dots, m_K)$ . The reranker learns scores  $g(s, m_i, \mathcal{C}_s)$  such that  $m^+$  is ordered as high as possible. Because it cannot return a molecule outside  $\mathcal{C}_s$ , end-to-end Recall@ $k$  is upper-bounded by

$$U_K = \frac{1}{N} \sum_{n=1}^N \mathbb{I}[m_n^+ \in \mathcal{C}_{s_n}]. \quad (2)$$

Define the within-list top- $k$  accuracy conditional on first-stage coverage as

$$A_{k|K} = \frac{\sum_{n=1}^N \mathbb{I}[m_n^+ \in \mathcal{C}_{s_n}, \text{rank}_g(m_n^+) \leq k]}{\sum_{n=1}^N \mathbb{I}[m_n^+ \in \mathcal{C}_{s_n}]}, \quad \text{Recall@}k = U_K A_{k|K} \leq U_K. \quad (3)$$

A reranker applied to a fixed cache can therefore change only  $A_{k|K}$ , not  $U_K$ . We report metrics over all queries, assigning zero credit when the positive is absent rather than conditioning the primary metric on successful first-stage retrieval.



**Fig. 1.** Overview of SpecEmbedding-Rerank. Panel 1 aligns the spectrum Transformer and molecular GINE. Panel 2 uses their frozen, cached embeddings to produce the top-40 list. Panel 3 applies shared pair features with either pointwise scoring or candidate-set self-attention and adds a residual to the base score. Dashed arrows denote training-only operations. Positives are never forced into the truncated top-40 list at validation/test, and no molecule is generated.

## 4 Method

### 4.1 Overview

The framework has three steps: cross-modal alignment, top- $K$  retrieval, and supervised residual reranking. Spectrum and molecule encoders are frozen while the reranker is trained, so their embeddings can be computed once and reused.

### 4.2 Cross-Modal Base Retriever

The spectrum encoder follows SpecEmbedding’s peak-sequence architecture [23], which processes peak  $m/z$  values and intensities with sinusoidal  $m/z$  features and a Transformer. In our implementation, each spectrum has at most 100 tokens: a precursor- $m/z$  token with intensity 2, followed by up to 99 fragments selected by intensity. Selected fragments retain their source-array order, and their intensities are normalized by the largest selected fragment intensity. Only these  $m/z$ -intensity tokens enter the spectrum encoder; adduct, instrument, and collision-energy metadata are not explicit inputs. We initialize this tower from scratch and train it against molecules rather than spectra. The molecular encoder is a GINE network over atom and bond features [11]. Mean-pooled graph features are augmented with explicit graph-size features. Two projection heads map the encoders to a common  $d$ -dimensional space:

$$\mathbf{z}_s = \frac{p_s(f_s(s))}{\|p_s(f_s(s))\|_2}, \quad \mathbf{z}_m = \frac{p_m(f_m(m))}{\|p_m(f_m(m))\|_2}. \quad (4)$$

We use a bidirectional cross-modal contrastive loss that permits multiple positives. Spectra and molecules sharing a molecular label form positives, which avoids treating replicate spectra as false negatives. For a minibatch of size  $B$ , let  $P(i)$  be the indices sharing the label of item  $i$ . With a learned inverse-temperature scale  $a$ , the base logits are

$$\ell_{ij} = a \mathbf{z}_{s_i}^\top \mathbf{z}_{m_j}. \quad (5)$$

The spectrum-to-molecule term is

$$\mathcal{L}_{s \rightarrow m} = -\frac{1}{B} \sum_{i=1}^B \frac{1}{|P(i)|} \sum_{j \in P(i)} \log \frac{\exp(\ell_{ij})}{\sum_{k=1}^B \exp(\ell_{ik})}, \quad (6)$$

and the molecule-to-spectrum term applies the same expression to  $\ell^\top$ . The alignment loss is their mean. The implementation parameterizes  $a = \exp(u)$  and caps it at 100. At inference, the normalized dot product  $b_i = \mathbf{z}_s^\top \mathbf{z}_{m_i}$  is the base score.

### 4.3 Candidate Construction and Training Protocol

MassSpecGym supplies one candidate pool per query. The base retriever scores the pool and keeps its top  $K = 40$  molecules. If a training target is absent from the retrieved top-40, it replaces the last molecule while its full-pool base rank is cached. That rank is one plus the number of strictly higher-scoring candidates; ties with the target do not count as higher. The rank embedding receives  $\bar{r}_i = \min\{\max(r_i, 0), 512\}$ , with index 0 reserved for padding. Training also shuffles candidate order while moving each candidate’s embedding, score, and rank together to guard against cache-order artifacts.

Scored pools are exact-SMILES-deduplicated unions of the labeled target and supplied candidates. A static audit found the target already present in all 32,010 frozen pools, so no pool was enlarged. Post-retrieval positive forcing is training-only; validation and test perform no top-40 replacement. A target absent from the retrieved top-40 remains absent and lowers the upper bound and all end-to-end metrics. No test record, target, or label is used for optimization or checkpoint selection. When forcing is applied, the inserted positive has a cached rank greater than 40, while the other 39 training candidates have ranks at most 40. Thus the rank embedding exposes a training-only label-correlated rank shortcut and creates a train-evaluation list-distribution shift. This is not test leakage: validation and test never perform the replacement. The no-rank-embedding audit below is therefore interpreted as a disclosure of this shortcut risk, not as evidence that rank has an independent positive effect.

### 4.4 Residual Reranker Variants

For candidate  $m_i$  with base rank  $r_i$ , we construct

$$\mathbf{x}_i = [\mathbf{z}_s, \mathbf{z}_{m_i}, \mathbf{z}_s \odot \mathbf{z}_{m_i}, |\mathbf{z}_s - \mathbf{z}_{m_i}|, b_i, \mathbf{e}(\bar{r}_i)], \quad (7)$$

where  $\mathbf{e}(\bar{r}_i)$  is a learned rank embedding. The direct embeddings retain modality-specific evidence, while product and absolute-difference terms make symmetric matching cues explicit. An MLP projects each vector:

$$\mathbf{h}_i = \text{MLP}(\mathbf{x}_i). \quad (8)$$

Both variants receive only the frozen spectrum embedding, frozen candidate embeddings, cosine base scores, clipped base-rank embeddings, and the candidate padding mask. Raw peaks, SMILES tokens, precursor mass or formula, and acquisition metadata are not passed directly to the reranker. Mass and formula constraints affect candidate-pool construction only; no explicit candidate-candidate structural-similarity feature is provided.

The candidate-set Transformer uses a Transformer encoder without candidate positional encodings to exchange information across candidates [22]:

$$[\tilde{\mathbf{h}}_1, \dots, \tilde{\mathbf{h}}_K] = \text{Transformer}([\mathbf{h}_1, \dots, \mathbf{h}_K]). \quad (9)$$

Padding masks support variable-sized lists. At deterministic inference, jointly permuting the candidates and their attached embeddings, scores, and ranks permutes the output scores in the same way; the base rank is a candidate-attached retrieval prior, not a tensor-slot position encoding [15,17]. The pointwise control sets  $\tilde{\mathbf{h}}_i = \mathbf{h}_i$ , retaining the same features, score head, residual path, and training losses while removing learned cross-candidate interaction after the retrieval priors have been constructed. It is not parameter- or capacity-matched because it removes the two Transformer layers: the pointwise and candidate-set variants contain exactly 1,478,690 and 7,783,458 trainable reranker parameters, respectively. A score head predicts a correction  $\Delta_i$ , and the final score is residual:

$$q_i = \alpha b_i + \Delta_i, \quad (10)$$

where  $\alpha$  is initialized to 1.0 and learned without a sign constraint. The residual path provides a direct shortcut from the base score; it does not guarantee preservation of the base order. Moreover, because  $b_i$  also enters  $\mathbf{x}_i$  and  $\Delta_i$  can learn additional score-dependent effects,  $\alpha$  alone is not a mechanistically interpretable measure of reliance on the base retriever.

#### 4.5 Learning Objective

For the labeled queries  $\mathcal{B}_L$  in a minibatch, let  $V_n$  be the valid candidate set and  $y_n$  the positive index. The primary objective averages listwise cross-entropy over labeled queries:

$$\mathcal{L}_{\text{list}} = -\frac{1}{|\mathcal{B}_L|} \sum_{n \in \mathcal{B}_L} \log \frac{\exp(q_{ny_n})}{\sum_{i \in V_n} \exp(q_{ni})}. \quad (11)$$

We also contrast each positive with every other valid candidate retrieved for that query. The implementation averages over all valid positive-negative pairs in the minibatch:

$$\mathcal{L}_{\text{pair}} = \frac{\sum_{n \in \mathcal{B}_L} \sum_{i \in V_n \setminus \{y_n\}} \log(1 + \exp(q_{ni} - q_{ny_n} + \gamma))}{\sum_{n \in \mathcal{B}_L} (|V_n| - 1)}. \quad (12)$$

The complete objective is

$$\mathcal{L} = \mathcal{L}_{\text{list}} + \lambda \mathcal{L}_{\text{pair}}. \quad (13)$$

The listwise term is a top-one likelihood: it separates the positive from the valid list but does not prescribe a total order among negatives. Both terms are invariant to adding the same query-specific constant to every  $q_i$ . Thus, the raw outputs are relative scores, not probabilities; probability calibration is neither performed nor evaluated.

For  $L$  Transformer layers and hidden width  $h$ , the candidate-set Transformer adds  $O(L(Kh^2 + K^2h))$  computation and  $O(LK^2)$  attention memory over the pointwise scorer. Neither variant decodes molecules token by token. An empirical latency comparison with GLMR is not reported, so we make no measured efficiency claim.

## 5 Experiments

### 5.1 Dataset and Candidate Pools

We use the published MassSpecGym split [3]: 194,119 training spectra, 19,429 validation spectra, and 17,556 test spectra. These counts total 231,104 spectra; the benchmark reports approximately 29,000 unique molecular structures. Its similarity-aware, structure-disjoint construction reduces close structural overlap between training and test molecules.

An exact audit of spectrum-record signatures (peak arrays, precursor  $m/z$ , adduct, instrument, and collision energy) found no target-molecule duplicates across splits, but six validation and three test signatures exactly matched training signatures. In all nine matched pairs, the target molecules differ and each evaluation query is marked `simulation_challenge=true`. We report these cases as cross-split spectrum-record overlap. In the overlap-clean alignment-seed-42 pipeline used as our canonical configuration, all validation-dependent model selection excludes the six affected validation spectra, leaving 19,423 validation queries for the rerankers; alignment validation keys decrease from 3,386 to 3,384 molecular entries. We retain and evaluate all 17,556 published test queries.

MassSpecGym supplies mass- and formula-conditioned candidate settings. The mass pool uses a precursor-mass tolerance; the formula pool assumes a known formula and is not unrestricted annotation. Frozen pools contain at most 256 candidates. Before truncation, their test-query means are 253.88 and 165.75, respectively; 1 mass and 3,159 formula pools contain fewer than 40 candidates. After top-40 truncation, the means are 40.00 and 35.73.

Our cache and evaluator assign one positive by exact target-SMILES equality. In official code commit `4f501b3`,<sup>1</sup> the reference loader instead defaults to two-dimensional InChIKey equivalence and may label multiple positives. Our Recall and MRR therefore use a local exact-target-SMILES single-positive protocol on the supplied candidate files; the effect of the identity-rule difference is unquantified. The multiple-positive labels used during alignment are likewise cached raw exact-SMILES labels, not two-dimensional molecular-identity labels. The graph features do not explicitly encode atom chirality or bond stereochemistry, so a stereo variant can be treated as a negative locally even when a two-dimensional identity rule would merge it; this model-level consequence has not been quantified.

## 5.2 Implementation Details

The aligned model outputs 512-dimensional embeddings. Its spectrum Transformer has four layers, 16 heads, and width 512; the molecular GINE has four layers, hidden width 128, and dropout 0.2. No published SpecEmbedding checkpoint is loaded, so the stage that would freeze a pretrained spectrum tower is skipped; both towers are optimized end to end on the MassSpecGym training split. Alignment uses AdamW with batch size 128, learning rate and weight decay  $10^{-4}$ , cosine scheduling, gradient-norm clipping at 1, at most 100 epochs, and patience 5. Each epoch makes ten draws per molecular label, randomly selecting an associated spectrum. Spectrum and graph augmentation are independently enabled with probability 0.5. The former removes a floor-rounded 20% of fragment peaks below normalized intensity 0.3 and jitters intensities by  $\pm 15\%$ ; the latter applies 10% node and 10% directed edge-entry dropout. Validation disables augmentation but retains replicate-spectrum sampling under the fixed random stream. We designate the overlap-clean project-default seed-42 run and its caches as the canonical configuration. Seed 42 is fixed by the project default configuration and was frozen before the 43/44 sensitivity runs; we do not claim that it was selected by test performance. This run applies the validation exclusion above, selects epoch 16, and stops at epoch 21.

The reranker has hidden width 512, a 32-dimensional rank embedding, and dropout 0.1. The candidate-set Transformer has two Transformer layers with eight attention heads; the pointwise control removes both layers. We use AdamW with learning rate  $10^{-4}$ , batch size 16, at most 30 epochs, and early stopping on validation MRR with patience 5. The pairwise weight and margin are  $\lambda = 0.2$  and  $\gamma = 0.1$ . For the canonical controlled analysis, the alignment-seed-42 checkpoint and its overlap-clean caches are fixed while each reranker variant is trained with seeds 42, 43, and 44. These are three reranker training seeds; each also controls initialization, data-loader order, candidate shuffling, and dropout. Their sample standard deviations measure reranker variation conditional on that first stage. A separate cross-alignment sensitivity analysis applies the same reranker seeds to alignment seeds 42, 43, and 44 and first aggregates reranker seeds within

<sup>1</sup> MassSpecGym reference loader and label transform, accessed August 1, 2026.

each alignment. Its cross-alignment unit is therefore the alignment checkpoint ( $n = 3$ ), not nine flattened runs. The component audit remains limited to the canonical alignment-seed-42 caches and reranker seeds. Its five removals respectively disable only the residual shortcut while retaining the base score as MLP input; jointly disable the base-score feature and shortcut; zero the rank embedding while preserving input width; remove product and absolute-difference features while retaining self-attention; or set  $\lambda = 0$  for listwise-only training. All models are selected without test-set access and evaluated on one NVIDIA RTX 4090.

Before training-only positive forcing, top-40 base recall is 93.78% for mass and 90.42% for formula on the training split. Training lists are then completed to 100% labeled coverage as described above. No target is forced into the test top-40 lists, whose corresponding upper bounds are 83.88% and 89.26%.

### 5.3 Code, Data, and Artifact Availability

The anonymous supplement provides source inspection, locked environment files, automated tests, and a tiny synthetic CPU smoke test. It does not provide the MassSpecGym spectra, candidate pools, trained alignment or reranker checkpoints, reranker caches, logs, or private analysis artifacts; the smoke test therefore does not reproduce the scientific metrics. The two reported pools use separate mass and formula reranker trainings. The frozen candidate artifacts and their provenance are identified by the release manifest; the analysis records source commits `a2280d2` (full model), `8bd2ca4` (ablations), and configuration hash prefix `b6260dad043f` (the full hash is retained in the analysis manifest). The disclosed features include atom, bond, and graph-size descriptors; the alignment projection is 512-dimensional with temperature  $\tau = 0.07$ ; AdamW uses weight decay  $10^{-4}$  and gradient clipping 1.0; spectrum caches are FP32 and molecule caches FP16. The real-data entry point is `run_rerank_pipeline.py`, subject to access to the excluded data and artifacts.

### 5.4 Baselines and Metrics

The controlled baselines are the same aligned retriever without the second-stage ranker and a pointwise reranker that removes candidate self-attention while retaining all other inputs and objectives, but not the same parameter count. We separately show JESTR and GLMR values from Table 1 of the GLMR paper [25]. These external numbers are not a controlled comparison: JESTR and GLMR were not independently reproduced in our codebase, and their retrievers, candidate handling, identity/canonicalization rules, and MCES implementations are not matched to our local exact-target-SMILES single-positive protocol.

We report Recall@1, Recall@5, Recall@20, and mean reciprocal rank (MRR) over all test queries. For every query, including those whose target is absent from the retrieved top-40, let  $\hat{m}_n^{(1)}$  be the top-ranked candidate. We report

**Table 2.** Project-default local MassSpecGym results. The base uses alignment seed 42, and learned rows report mean  $\pm$  sample SD over reranker training seeds 42–44. Recall and MRR are percentages under our exact-target-SMILES single-positive protocol on supplied candidate files; they are not official-evaluator-equivalent. MCES is reported separately as a single-checkpoint audit.

| Pool    | Method                                           | R@1              | R@5              | R@20             | MRR              |
|---------|--------------------------------------------------|------------------|------------------|------------------|------------------|
| Mass    | SpecMolAlign (alignment seed 42)                 | 47.46            | 63.51            | 77.76            | 55.03            |
|         | Pointwise reranker (training seeds 42–44)        | 67.44 $\pm$ 0.74 | 77.02 $\pm$ 0.59 | 82.26 $\pm$ 0.16 | 71.79 $\pm$ 0.66 |
|         | Candidate-set Transformer (training seeds 42–44) | 68.58 $\pm$ 0.20 | 77.36 $\pm$ 0.36 | 82.20 $\pm$ 0.07 | 72.60 $\pm$ 0.25 |
| Formula | SpecMolAlign (alignment seed 42)                 | 63.10            | 75.79            | 84.93            | 68.86            |
|         | Pointwise reranker (training seeds 42–44)        | 73.97 $\pm$ 0.59 | 81.94 $\pm$ 0.27 | 86.78 $\pm$ 0.27 | 77.64 $\pm$ 0.49 |
|         | Candidate-set Transformer (training seeds 42–44) | 74.57 $\pm$ 0.28 | 81.97 $\pm$ 0.26 | 86.77 $\pm$ 0.14 | 78.02 $\pm$ 0.26 |

**Table 3.** Single-checkpoint thresholded myopic-MCES@1 audit on the project-default alignment-seed-42 checkpoint and reranker training seed 42; lower is better. The solver was PULP\_CBC\_CMD; zero failed pairs were logged. No seed-level mean or standard deviation is implied.

| Pool    | Base    | Candidate-set | Transformer |
|---------|---------|---------------|-------------|
| Mass    | 15.3681 |               | 7.7057      |
| Formula | 5.4389  |               | 3.0913      |

$$\text{MCES@1} = \frac{1}{N} \sum_{n=1}^N \tilde{d}_{15}(\hat{m}_n^{(1)}, m_n^+), \quad N = 17,556. \quad (14)$$

Here  $\tilde{d}_{15}$  is the output of **myopic-MCES** with threshold 15 and the stronger-bound option enabled [14]. It is exact for sufficiently close pairs; for distant pairs it may return the threshold or a computed lower bound rather than the unconstrained exact MCES distance. Thus, the reported quantity is thresholded myopic MCES, and lower is better. Exact SMILES-string matches are assigned zero. The canonical runs selected PULP\_CBC\_CMD, and no MCES worker failures were logged. Recall and MRR seed dispersion is computed over three reranker seeds with a fixed base checkpoint in the canonical audit. The cross-alignment summary first averages reranker seeds within each checkpoint and then uses the three alignment-level paired estimates; neither sample standard deviation is a confidence interval. MCES@1 is computed for the overlap-clean base and project-default seed-42 Transformer checkpoints; these single-checkpoint values do not include seed-level uncertainty.

## 5.5 Main Results

In the representative seed-42 run, the candidate-set Transformer adds 21.28 percentage points in mass-conditioned Recall@1 and 11.65 points in formula-conditioned Recall@1. The corresponding MRR gains are 17.84 and 9.36 points.

**Table 4.** Alignment-stratified internal results using MassSpecGym supplied-candidate top-40 lists and our local exact-target-SMILES single-positive evaluation, which is not official-evaluator-equivalent. Learned entries are means over reranker seeds 42–44 within each alignment. The three checkpoints come from two code revisions and provide descriptive sensitivity evidence, not confidence intervals, significance tests, or general end-to-end random-seed stability. Recall@1 and MRR are percentages.

| Alignment Pool |         | U40 Base | R@1   | Point. | R@1   | Trans. | R@1   | Base MRR | Point. | MRR | Trans. | MRR |
|----------------|---------|----------|-------|--------|-------|--------|-------|----------|--------|-----|--------|-----|
| 42             | Mass    | 83.88    | 47.46 | 67.44  | 68.58 | 55.03  | 71.79 | 72.60    |        |     |        |     |
| 42             | Formula | 89.26    | 63.10 | 73.97  | 74.57 | 68.86  | 77.64 | 78.02    |        |     |        |     |
| 43             | Mass    | 80.98    | 35.92 | 66.92  | 67.19 | 44.33  | 70.42 | 70.58    |        |     |        |     |
| 43             | Formula | 81.65    | 45.08 | 69.64  | 69.30 | 51.85  | 72.14 | 71.96    |        |     |        |     |
| 44             | Mass    | 72.20    | 23.91 | 57.78  | 57.42 | 32.26  | 61.22 | 60.91    |        |     |        |     |
| 44             | Formula | 75.60    | 28.73 | 59.76  | 59.82 | 36.87  | 63.14 | 63.34    |        |     |        |     |

**Table 5.** Controlled comparison on fixed alignment-seed-42 caches. The base row is fixed; learned rows report mean  $\pm$  sample standard deviation over reranker seeds 42–44. Recall@1 and MRR are percentages. Removal rows are separately trained audits of the predefined full configuration, not post-hoc model-selection candidates. All values use the local exact-target-SMILES single-positive protocol and are not official-evaluator-equivalent.

| Method                              | Mass             |                  | Formula          |                  |
|-------------------------------------|------------------|------------------|------------------|------------------|
|                                     | R@1              | MRR              | R@1              | MRR              |
| Base                                | 47.46            | 55.03            | 63.10            | 68.86            |
| Pointwise                           | 67.44 $\pm$ 0.74 | 71.79 $\pm$ 0.66 | 73.97 $\pm$ 0.59 | 77.64 $\pm$ 0.49 |
| Candidate-set Transformer (full)    | 68.58 $\pm$ 0.20 | 72.60 $\pm$ 0.25 | 74.57 $\pm$ 0.28 | 78.02 $\pm$ 0.26 |
| No residual shortcut                | 68.16 $\pm$ 0.06 | 72.33 $\pm$ 0.06 | 75.01 $\pm$ 0.54 | 78.38 $\pm$ 0.42 |
| No base-score input/shortcut        | 68.07 $\pm$ 0.20 | 72.28 $\pm$ 0.18 | 74.39 $\pm$ 0.55 | 77.93 $\pm$ 0.38 |
| No rank embedding                   | 70.12 $\pm$ 0.30 | 73.75 $\pm$ 0.28 | 76.62 $\pm$ 0.69 | 79.38 $\pm$ 0.58 |
| Product/absolute-difference removed | 68.10 $\pm$ 0.85 | 72.29 $\pm$ 0.66 | 75.76 $\pm$ 0.55 | 78.90 $\pm$ 0.38 |
| Listwise only ( $\lambda = 0$ )     | 68.91 $\pm$ 0.76 | 72.86 $\pm$ 0.46 | 74.24 $\pm$ 0.25 | 77.71 $\pm$ 0.25 |

Improvements at Recall@20 are smaller because the first-stage top-40 upper bound is already the dominant constraint. Relative to the corresponding fixed overlap-clean base, the same single-checkpoint seed-42 Transformer in Table 3 reduces MCES@1 from 15.3681 to 7.7057 for mass candidates (49.86%) and from 5.4389 to 3.0913 for formula candidates (43.16%). These are single-checkpoint audits, not seed-aggregated estimates.

JESTR and GLMR remain reported-only, not reproduced, and not directly comparable background methods. We retain their mechanism descriptions but do not reproduce their numeric table or infer a relative ranking.

## 5.6 Cross-Alignment and Controlled Component Evidence

Table 4 preserves the alignment hierarchy: reranker seeds are first averaged within each first-stage checkpoint. Both learned rankers improve the alignment-matched base in Recall@1 and MRR in all 12 alignment-candidate-learned-model aggregate cells (three alignments  $\times$  two candidate pools  $\times$  two learned

models; each cell improves both Recall@1 and MRR). These are aggregate cells, not independent reranker runs. In contrast, the Transformer–pointwise difference is positive for only two of three alignments for each candidate setting and metric. Across the three alignment-level paired estimates, the mean  $\pm$  sample-SD differences are  $0.35 \pm 0.75$  Recall@1 points and  $0.22 \pm 0.56$  MRR points for mass, and  $0.11 \pm 0.47$  and  $0.13 \pm 0.29$  points for formula. These are descriptive summaries, not confidence intervals or significance tests. The three checkpoints share the same recorded hyperparameters but were produced from two code revisions; per-alignment provenance is retained in the analysis artifacts. Thus, these internal results support supervised residual reranking but do not establish a checkpoint-robust independent benefit of candidate self-attention.

On the canonical alignment-seed-42 checkpoint in Table 5, the candidate-set Transformer exceeds the pointwise model by 1.14/0.60 Recall@1 points and 0.81/0.38 MRR points for mass/formula candidates, respectively. For each of the six pool-by-reranker-seed pairs (two candidate pools  $\times$  three reranker training seeds), both Recall@1 and MRR favor the Transformer within this checkpoint. Because it also has 6.30M more trainable parameters, even this within-checkpoint comparison does not isolate candidate interaction from increased capacity; the cross-alignment direction changes further preclude a checkpoint-robust self-attention claim.

The separately trained removal rows audit the full configuration on the same fixed alignment. The full model does not achieve six-of-six MRR wins against any removal variant. Removing the rank embedding improves both reported metrics in every pool-by-reranker-seed pair; its mean Recall@1/MRR gains are 1.54/1.15 mass points and 2.05/1.36 formula points. The residual shortcut, product/absolute-difference features, and pairwise loss have candidate-dependent directions. The full model’s mean MRR lead over the joint base-score-pathway removal is only 0.32/0.08 points, with mixed seed signs. These are descriptive comparisons, not significance tests; the predefined full configuration remains the reporting reference.

For auditability, the six-of-six MRR win counts (positive/tied/negative, abbreviated P/Z/N) over the two candidate pools  $\times$  three reranker training seeds are: no residual shortcut 2/0/4, no base-score pathway 3/0/3, no rank embedding 0/0/6, no interaction features 2/0/4, and listwise-only training 3/0/3. These are six paired reranker comparisons per removal, not independent alignment replications.

On the pre-overlap-clean fixed checkpoint, a sensitivity check excluded the three affected test queries, leaving 17,553. Across both candidate settings and rerankers, mean Recall@1 decreased by 0.0045–0.0055 points (maximum seed-level change: 0.0056); all changes were below 0.006 percentage points. Canonical tables retain all test queries.

The overlap-clean and pre-clean pipelines select the same alignment epochs (best 16, stop 21), but reranker best/stop epochs differ in 9 of 12 configurations. Because alignment was retrained and its test metrics changed, this measures pipeline sensitivity rather than the causal effect of six validation queries.

We do not report candidate-order or full top- $K$  ablations. We also do not report peak memory, cache construction time, reranker latency, or end-to-end latency.

## 6 Discussion

Equations 2–3 separate first-stage coverage from within-list ordering. For  $k = 1$  and  $K = 40$ , the defined conditional quantity  $A_{1|40}$  equals  $R@1/U_{40}$ . Using the displayed three-seed means, it increases from 56.58% for the base retriever to 80.40% for the pointwise model and 81.76% for the candidate-set Transformer under mass conditioning. The corresponding formula-conditioned rates are 70.69%, 82.87%, and 83.54%.

Another view is the fraction of attainable top-1 headroom recovered,

$$H = \frac{R@1_{\text{rerank}} - R@1_{\text{base}}}{U_{40} - R@1_{\text{base}}}. \quad (15)$$

The pointwise and candidate-set models recover 54.86% and 57.99% of this headroom for mass candidates, and 41.55% and 43.85% for formula candidates. The pointwise gain is also 94.60% and 94.77% of the total candidate-set gain over the base model in the two settings. Thus, most of the observed improvement is already achieved by the supervised pointwise second-stage ranker as configured; the candidate-set variant adds only a modest increment. At Recall@20, the candidate-set three-seed means of 82.20% and 86.77% are only 1.68 and 2.49 points below  $U_{40}$ ; first-stage coverage is therefore the dominant remaining limitation for that metric under this top-40 protocol. The removals likewise support only a framework-level interpretation: the full model does not achieve six-of-six MRR wins against any removal variant, whereas rank-embedding removal improves both metrics in every pool-by-reranker-seed pair. The predefined full configuration remains the reporting reference; these algebraic and removal comparisons are not causal estimates of individual components. Across alignment seeds 42–44, both learned variants retain a large descriptive margin over their corresponding base, despite substantial first-stage metric variation. This strengthens the framework-level observation within the three audited runs, while the mixed Transformer–pointwise directions reinforce that most of the gain should not be attributed to candidate self-attention.

## 7 Limitations and Responsible Evaluation

Three-seed sample standard deviations and paired-win counts are descriptive, not confidence intervals or significance tests. The canonical table and all component removals fix alignment seed 42. The separate cross-alignment analysis covers only three internal checkpoints, produced by two code revisions, and first aggregates reranker seeds within each alignment; it is descriptive sensitivity evidence rather than a confidence interval or general stability claim. The component audit therefore establishes neither cross-alignment component stability

nor causal effects. The joint base-score-pathway removal changes both its MLP input and shortcut; product/absolute-difference removal retains self-attention; and the no-rank-embedding variant was neither tested across alignments nor selected post hoc. The candidate-set model also has 5.26 times the pointwise model’s trainable parameters (6,304,768 more; +426%), confounding interaction with capacity. External JESTR and GLMR rows were not reproduced under identical software and candidate handling; we claim neither definitive state of the art nor an independent component benefit.

Training-only positive replacement changes the list distribution without addressing  $1 - U_K$ . Because frozen full pools contain the exact target, pool-level evaluation is closed-set even when top-40 retrieval misses it. The impact of exact-target-SMILES rather than reference two-dimensional InChIKey identity is unquantified, so Recall/MRR are explicitly local-protocol results rather than official-evaluator-equivalent values. The exact-record audit cannot exclude near duplicates or correlated provenance. Model selection excludes the six audited training-validation overlaps; all test queries remain. The post-hoc validation environment is now version-locked and was rebuilt from explicit Conda/PyPI locks, but historical training runs did not preserve a per-package environment record; therefore we do not claim bitwise retraining equivalence. A 2026 audit preprint reports further MassSpecGym leakage, shortcut, implementation, and metric risks and releases v1.5 [16]; our scores are artifact- and protocol-specific, not proven v1.5-equivalent.

Alignment sampling approximately balances molecular labels, whereas metrics count spectra, giving replicate-rich molecules more weight. Candidate-pool Recall measures prioritization, not confirmed identification; confidence frameworks and work on spectral-library absences distinguish these notions [19,10]. Conformal retrieval can produce spectrum-specific sets with marginal coverage under exchangeability and target-in-pool assumptions [18], but such guarantees do not automatically cover a target missing from the retrieved top-40. We perform neither probability calibration nor abstention; orthogonal experimental evidence remains necessary.

Finally, formula conditioning presumes a known formula. The spectrum tower uses selected peaks and precursor  $m/z$ , but no explicit adduct, instrument, or collision-energy inputs. Robustness to acquisition shift, other retrievers and embeddings, identity rules, pool construction, larger lists, other benchmarks, and external libraries is untested. The closed-set method cannot propose structures outside the supplied pool despite avoiding autoregressive generation-validity failures.

## 8 Conclusion

We presented SpecEmbedding-Rerank for supervised, non-generative second-stage reranking of MS/MS candidates. Recall and MRR use MassSpecGym-supplied candidate files with a local exact-target-SMILES single-positive protocol. Because the reference loader defaults to two-dimensional InChIKey equiva-

lence and may yield multiple positives, the unquantified difference means these results are not official-evaluator-equivalent. Within this protocol, both learned variants improve their alignment-matched base in both metrics in all 12 aggregate cells across alignment seeds 42–44. The candidate-set Transformer has a modest canonical-checkpoint advantage over the pointwise scorer, but its direction is positive in only two of three alignments for each candidate setting and metric. The full model does not achieve six-of-six MRR wins against any of five removal variants within the canonical alignment, while rank-embedding removal improves both reported metrics in every pool-by-reranker-seed pair. The strongest supported conclusion therefore concerns the reranking framework, not an independent component benefit. The separately displayed JESTR/GLMR values were not reproduced and are not directly comparable; matched external baselines and measured efficiency remain outside the evidence.

## References

1. Bach, E., Schymanski, E.L., Rousu, J.: Joint structural annotation of small molecules using liquid chromatography retention order and tandem mass spectrometry data. *Nature Machine Intelligence* **4**(12), 1224–1237 (2022). <https://doi.org/10.1038/s42256-022-00577-2>
2. Burges, C., Shaked, T., Renshaw, E., Lazier, A., Deeds, M., Hamilton, N., Hullender, G.: Learning to rank using gradient descent. In: *Proceedings of the 22nd International Conference on Machine Learning*. pp. 89–96 (2005). <https://doi.org/10.1145/1102351.1102363>
3. Bushuiev, R., Bushuiev, A., de Jonge, N.F., Young, A., Kretschmer, F., Samusevich, R., Heirman, J., Wang, F., Zhang, L., Dührkop, K., Ludwig, M., Haupt, N.A., Kalia, A., Brungs, C., Schmid, R., Greiner, R., Wang, B., Wishart, D.S., Liu, L.P., Rousu, J., Bittremieux, W., Röst, H., Mak, T.D., Hassoun, S., Huber, F., van der Hooft, J.J.J., Stravs, M.A., Böcker, S., Sivic, J., Pluskal, T.: MassSpecGym: A benchmark for the discovery and identification of molecules. In: *Advances in Neural Information Processing Systems*. vol. 37, pp. 110010–110027 (2024). <https://doi.org/10.52202/079017-3491>
4. Cao, Z., Qin, T., Liu, T.Y., Tsai, M.F., Li, H.: Learning to rank: From pairwise approach to listwise approach. In: *Proceedings of the 24th International Conference on Machine Learning*. pp. 129–136 (2007). <https://doi.org/10.1145/1273496.1273513>
5. Chen, L., Xia, B., Wang, Y., Huang, X., Gu, Y., Wu, W., Zhou, Y.: CMSSP: A contrastive mass spectra-structure pretraining model for metabolite identification. *Analytical Chemistry* **96**(42), 16871–16881 (2024). <https://doi.org/10.1021/acs.analchem.4c03724>
6. Chen, Y.Z., Hassoun, S.: Learning from all views: A multiview contrastive framework for metabolite annotation. *Analytical Chemistry* **98**(9), 6598–6606 (2026). <https://doi.org/10.1021/acs.analchem.5c05675>
7. Dührkop, K., Shen, H., Meusel, M., Rousu, J., Böcker, S.: Searching molecular structure databases with tandem mass spectra using CSI:FingerID. *Proceedings of the National Academy of Sciences* **112**(41), 12580–12585 (2015). <https://doi.org/10.1073/pnas.1509788112>

8. Gerlich, M., Neumann, S.: MetFusion: Integration of compound identification strategies. *Journal of Mass Spectrometry* **48**(3), 291–298 (2013). <https://doi.org/10.1002/jms.3123>
9. Goldman, S., Wohlwend, J., Stražar, M., Haroush, G., Xavier, R.J., Coley, C.W.: Annotating metabolite mass spectra with domain-inspired chemical formula transformers. *Nature Machine Intelligence* **5**(9), 965–979 (2023). <https://doi.org/10.1038/s42256-023-00708-3>
10. Hoffmann, M.A., Nothias, L.F., Ludwig, M., Fleischauer, M., Gentry, E.C., Witting, M., Dorrestein, P.C., Dührkop, K., Böcker, S.: High-confidence structural annotation of metabolites absent from spectral libraries. *Nature Biotechnology* **40**(3), 411–421 (2022). <https://doi.org/10.1038/s41587-021-01045-9>
11. Hu, W., Liu, B., Gomes, J., Zitnik, M., Liang, P., Pande, V., Leskovec, J.: Strategies for pre-training graph neural networks. In: *International Conference on Learning Representations* (2020), <https://openreview.net/forum?id=HJlWWJSFDH>
12. de Jonge, N.F., Louwen, J.J.R., Chekmeneva, E., Camuzeaux, S., Vermeir, F.J., Jansen, R.S., Huber, F., van der Hooft, J.J.J.: MS2Query: Reliable and scalable MS2 mass spectra-based analogue search. *Nature Communications* **14**(1), 1752 (2023). <https://doi.org/10.1038/s41467-023-37446-4>
13. Kalia, A., Chen, Y.Z., Krishnan, D., Hassoun, S.: JESTR: Joint embedding space technique for ranking candidate molecules for the annotation of untargeted metabolomics data. *Bioinformatics* **41**(7), btaf354 (2025). <https://doi.org/10.1093/bioinformatics/btaf354>
14. Kretschmer, F., Seipp, J., Ludwig, M., Klau, G.W., Böcker, S.: Coverage bias in small molecule machine learning. *Nature Communications* **16**, 554 (2025). <https://doi.org/10.1038/s41467-024-55462-w>
15. Lee, J., Lee, Y., Kim, J., Kosiorek, A., Choi, S., Teh, Y.W.: Set transformer: A framework for attention-based permutation-invariant neural networks. In: *Proceedings of the 36th International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 97, pp. 3744–3753. PMLR (2019), <https://proceedings.mlr.press/v97/lee19d.html>
16. Liu, H., Bushuiev, R., Lightheart, I., Manjrekar, M., Bushuiev, A., Lederbauer, M., Jozefov, F., Wang, Y., Hassoun, S., Sivic, J., Taylor, J., Wang, R., Healey, D., Pluskal, T., Coley, C.W.: MassSpecGym in the wild: Uncovering and correcting evaluation pitfalls in AI-driven molecule discovery (2026). <https://doi.org/10.48550/arXiv.2606.19624>, preprint
17. Pang, L., Xu, J., Ai, Q., Lan, Y., Cheng, X., Wen, J.: SetRank: Learning a permutation-invariant ranking model for information retrieval. In: *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*. pp. 499–508 (2020). <https://doi.org/10.1145/3397271.3401104>
18. Rakhshaninejad, M., De Waele, G., Jürgens, M., Waegeman, W.: Reliable molecular retrieval from mass spectra using conformal prediction. *Journal of Chemical Information and Modeling* **66**(10), 5788–5800 (2026). <https://doi.org/10.1021/acs.jcim.6c00727>
19. Schymanski, E.L., Jeon, J., Gulde, R., Fenner, K., Ruff, M., Singer, H.P., Hollender, J.: Identifying small molecules via high resolution mass spectrometry: Communicating confidence. *Environmental Science & Technology* **48**(4), 2097–2098 (2014). <https://doi.org/10.1021/es5002105>
20. Stravs, M.A., Dührkop, K., Böcker, S., Zamboni, N.: MSNovelist: De novo structure generation from mass spectra. *Nature Methods* **19**(7), 865–870 (2022). <https://doi.org/10.1038/s41592-022-01486-3>

21. Tian, Y., Krishnan, D., Isola, P.: Contrastive multiview coding. In: Computer Vision – ECCV 2020. Lecture Notes in Computer Science, vol. 12356, pp. 776–794. Springer, Cham (2020). [https://doi.org/10.1007/978-3-030-58621-8\\_45](https://doi.org/10.1007/978-3-030-58621-8_45)
22. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I.: Attention is all you need. In: Advances in Neural Information Processing Systems. vol. 30, pp. 5998–6008 (2017), <https://proceedings.neurips.cc/paper/7181-attention-is-all-you-need>
23. Xiong, P., Xu, H., Zheng, H.: Supervised contrastive learning leads to more reasonable spectral embeddings. *Analytical Chemistry* **97**(37), 20137–20146 (2025). <https://doi.org/10.1021/acs.analchem.5c02655>
24. Young, A., Röst, H., Wang, B.: Tandem mass spectrum prediction for small molecules using graph transformers. *Nature Machine Intelligence* **6**(4), 404–416 (2024). <https://doi.org/10.1038/s42256-024-00816-8>
25. Zhang, Y., Ding, K., Wu, Y., Zhuang, X., Yang, Y., Zhang, Q., Chen, H.: Breaking the modality barrier: Generative modeling for accurate molecule retrieval from mass spectra. *Proceedings of the AAAI Conference on Artificial Intelligence* **40**(2), 1561–1569 (2026). <https://doi.org/10.1609/aaai.v40i2.37132>